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Project Phase 4

Phase Introduction:

There have been many new CNN architectures that have been proposed in the last 10 years that deeply strengthened their performance for image classification. However, when trying to decide which models to look at I wanted to keep in mind the main design constraint I imposed on this project; minimizing model size (parameter count) while maximizing performance. That design constraint combined with their online availability for comparison is why I chose these two specific models to reference;

- ConvNeXt-V2 from Meta AI [1]
- MobileNets from Google Research [2]

While MobileNets pioneered the use of depthwise separable convolutions for lightweight vision, ConvNeXt-V2 represents the modern "state-of-the-art" for pure CNNs, modernizing the convolutional block to compete with Vision Transformers. Though Vision transformers may truly be a more state of the art architecture I wanted to focus here on looking at state of the art CNN architectures.

Both of these networks were evaluated on the benchmark Image Net dataset; the full ImageNet project contains over 14 million images across 21,000+ categories. However, researchers often use the ImageNet-1K subset for model evaluation.

- **Scale:** It contains approximately 1.28 million training images, 50,000 validation images, and 100,000 test images.
- **Classes:** There are 1,000 distinct object categories. These categories are based on the WordNet hierarchy and include a wide range of subjects, from specific dog breeds (fine-grained) to vehicles and household items.
- **Resolution:** Unlike CIFAR-100, ImageNet images are high-resolution and variable in size. However, for training models they are typically pre-processed (resized and center-cropped) to a fixed square resolution.

ConvNeXt-V2:

Core Innovation: ConvNeXt-V2 evolved from the original ConvNeXt by integrating a Fully Convolutional Masked Autoencoder (FCMAE) framework and a novel Global Response Normalization (GRN) layer. It proves that pure CNNs can achieve performance similar to that of Vision Transformers by adopting "Transformer-like" design choices, such as larger kernels (7x7), Layer Normalization, and inverted bottleneck blocks, without the complexity of self-attention mechanisms.

Fully Convolutional Masked Autoencoders: In this framework, large portions of an input image are randomly masked out, and the model is tasked with reconstructing the missing pixels based only on the visible patches. The fully convolutional aspect is critical because it utilizes sparse convolutions to process the image. Unlike standard convolutions that slide a window over every pixel (treating masks as "zero" values that still influence the math), sparse convolutions operate exclusively on the visible, unmasked coordinates. This prevents "information leakage," where data from unmasked regions would otherwise bleed into masked areas during the forward pass, and ensures the model doesn't waste parameters learning to identify the mask pattern itself. By treating masked regions as truly absent data, the CNN is forced to learn high-level spatial relationships and global structure to predict the missing content, effectively priming its internal weights for superior feature representation.

Global Response Normalization: The Global Response Normalization (GRN) layer is a feature-competition mechanism introduced in ConvNeXt-V2 to solve the problem of "feature collapse," where many neurons in a deep CNN become redundant or inactive. It functions by first calculating the L2-norm of each feature channel across its spatial dimensions to determine its "global response." This response is then normalized against the responses of all other channels in the same layer, creating a contrastive effect where strong, informative features are enhanced and weak, redundant ones are suppressed. Finally, the layer applies learnable scaling and bias parameters to integrate this normalized signal back into the original feature map.

Performance Summary: The smallest variant, ConvNeXt-V2 Atto, is specifically designed for efficiency. It achieves a Top-1 accuracy of 76.7% on ImageNet-1K with only 3.7M parameters [1].

<u>MobileNets:</u>
MobileNets are a class of small scale CNNs designed to be efficient for use in mobile device vision applications [2].

Core Innovation: Introduced by Howard et al. (Google), MobileNets are built primarily on depthwise separable convolutions to drastically reduce computation (mult-adds) and model size. It introduces two hyper-parameters: the Width Multiplier (α) and the Resolution Multiplier (ρ), allowing designers to trade off accuracy for latency and size. The depthwise separable convolutions also allow for the model to have more depth in the same amount of parameters as a model with only standard convolutional blocks.

Depthwise Separable Convolutions: Depthwise separable convolutions are a highly efficient form of convolutional operation that factorizes a standard convolution into two distinct sequential steps: a depthwise convolution and a pointwise convolution. In the depthwise step, a single spatial filter is applied independently to each input channel (filtering spatial correlations), while the pointwise step uses a 1x1 convolution to combine these filtered channels into a new channel space (filtering cross-channel correlations). By decoupling these two processes, the architecture drastically reduces the number of parameters and computational operations required.

For a convolutional layer with 32 channels, 3x3 kernel size, and 16 input channels the parameter count (ignoring bias) for a standard convolutional block would be (9 x 32 x 16) = 4,608 with a depthwise separable convolution with the same sizes the parameter count would be 16 x (9 + 32) = 656. This is a reduction of 7x.

Performance Summary: A standard V1 MobileNet achieves 70.6% Top-1 accuracy on ImageNet with only 4.2 million parameters [2]. A standard successor V2 MobileNet achieves 72% accuracy with 3.5 million parameters [3].

Numerical Comparison:
My model achieves 67.9% accuracy on the 100 class classification task with just 2.7 million parameters. This is 3% lower than the standard MobileNets achieve on the 1000 class classification task of the ImageNet-1k dataset. MobileNets also performed better on ImageNet than my model did on the Cifar-100 with fewer parameters. The .75 MobileNet (Width Multiplier = .75) reduces the parameter count from the standard 4.2 million to 2.6 million. With this model the MobileNet achieved 68.4% accuracy on ImageNet. The standard successor V2 MobileNet achieved 72% accuracy on the same dataset. However, its reduced model containing 2.6 million parameters returned 69.8% accuracy on ImageNet, ~1% better than the original version.

Side Note Hypothesis: Part of the ability of these nets to perform so well with so few parameters on a classification task that has 10x more classes (far more complexity) than my task has to do with the training set sizes. As was seen in the previous phase data augmentation was the most impactful (except batchnorm which caused divergence when removed) component to model performance. This is because it artificially inflates the training set size and diversity. The Cifar-100 has 50,000 training images (500 per class) 10,000 of which were held out for the validation set, these numbers are tiny compared to the size of the ImageNet dataset, which is, 1.28 million training images and 50,000 validation images.

The ConvNeXt-V2 models which are truly state of the art (benchmark performance) achieved 76.2% accuracy with their Atto model. This version of the model contains 3.7 million parameters and performs 10% worse than the version of the model with 660 million parameters. While this model is slightly larger (which is not the reason it performs better) than both my model and the MobileNet the improvement is massive. 76.2% is ~8% better than MobileNet on the same dataset and ~8% better than my model performed on the Cifar-100 dataset.

Implementation:

For the descriptive numerical comparison I selected MobileNet V2. I will load a MobileNet V2 model and adjust the hyperparameters (Width Multiplier) to ensure its parameter count matches my final model. Both models will then be trained and tested on the same dataset for a direct performance comparison.

I would like to load the ConvNeXt-V2 Atto model as it is the most state of the art CNN since a majority of research has shifted towards vision transformers in the past few years. However, like many modern architectures, it is typically pre-trained at 224x224 resolution with a 7x7 kernel. This means testing it on the native 32x32 CIFAR-100 dataset would require massive upsampling. There is a concern that this upsampling combined with the aggressive initial spatial reduction inherent in high-resolution architectures might yield results that are not truly reflective of the model's capabilities due to spatial dimension mismatch. This spatial dimension mismatch is not a problem with the MobileNets as their design leads them to natively (with minor changes) handle images as small as 32x32 comfortably.

References:
[1] S. Woo, S. Debnath, R. Hu, X. Chen, Z. Liu, N. Zhuang, and C. Feichtenhofer, "ConvNeXt V2: Co-designing and Scaling ConvNets with Masked Autoencoders," *arXiv preprint arXiv:2301.00808*, 2023. [Online]. Available: https://arxiv.org/pdf/2301.00808
[2] A. G. Howard *et al.*, "MobileNets: Efficient Convolutional Neural Networks for Mobile Vision Applications," *arXiv preprint arXiv:1704.04861*, 2017. [Online]. Available: https://arxiv.org/abs/1704.04861
[3] M. Sandler, A. Howard, M. Zhu, A. Zhmoginov, and L. C. Chen, "MobileNetV2: Inverted Residuals and Linear Bottlenecks," in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2018, pp. 4510-4520. [Online]. Available: https://arxiv.org/abs/1801.04381